COMP7015 Artificial Intelligence (26/27) – Group Project Instructions

1. Overall Requirements

1. **Groups:** Form a group of **1 to 6** students. Forming groups with students from other sections is allowed. (At least 3 members in a team are recommended)
2. **Milestones:**
   - **Group Registration & Topic Selection:** Due by 11:59 pm, 23rd October 2026. Please send an email to our TA, Mr. Pinzhen Chi (cspzchi@comp.hkbu.edu.hk), and copy all your group members in the email. Please include the student ID and full name of all members, as well as the topic that your group selects, in the email.
   - **Final Submission:** Due by 11:59 pm, 13th November 2026. This includes your source code and project report. One submission for each team will be enough.
   - **In-person Presentation:** Scheduled for 14th and 15th November 2026. A detailed schedule will be announced after the group registration deadline.
3. **Final Submission Package:**
   - **Project Report:** A PDF document of at most five A4 pages (single column). It should describe your project's motivation, methods, results, and a discussion. A mandatory section must detail the contribution of each group member.
   - **Source Codes:** A single .zip file. Your code will be evaluated in the FSC 8/F lab environment, so ensure it runs smoothly there. Acknowledge all major third-party libraries (e.g., PyTorch, TensorFlow, Hugging Face Transformers) in your report.
4. **Presentation:** Each group will have approximately 8 minutes for their presentation, followed by a Q&A session. Every member must present their part of the work. The use of visualizations (figures, graphs, demos) is highly encouraged to clearly convey your project's story.
5. **Academic Integrity:**
   - We have a zero-tolerance policy for plagiarism. All submissions will be checked by anti-plagiarism software. Copying code from online sources or generative AI tools without proper citation and understanding is strictly prohibited.
   - Submission of work from other courses or previous projects is considered self-plagiarism and is not allowed.

2. Topics

You may choose one of the following three topics. For Topics 1 and 2, you are expected to build on the suggested tasks. For Topic 3, you have the freedom to define your own project, keeping in mind that the project's scope and difficulty will be evaluated.

Topic 1: Plant Disease Classification (Computer Vision)

This project focuses on building and evaluating deep learning models for plant disease classification using the **PlantVillage dataset**. It contains over 54,000 images of healthy and diseased leaves from 14 crop species. The goal is to identify the crop and disease or healthy condition from a leaf image.

Dataset:

1. This topic is based on the PlantVillage dataset, available for public download at https://github.com/spMohanty/PlantVillage-Dataset
2. Use the original RGB images (raw/color) and the repository's leaf grouping metadata. Dataset details are also available at https://huggingface.co/datasets/mohanty/PlantVillage.
3. Select at least four classes from one crop, including healthy leaves and three diseases (e.g., tomato healthy, bacterial spot, early blight, and late blight). More classes or crops are optional; meeting this minimum is sufficient regardless of other groups' dataset sizes.

Minimum Requirements (you secure 50% scores if the following are done correctly):

1. Data split: Create training, validation, and test sets using the repository's leaf-group metadata. Keep all images and augmented versions of the same leaf in one split. Report class counts and the seed; use validation for tuning and the test set only for final evaluation.
2. Image preprocessing: Resize and normalize RGB images and verify labels. Apply symptom-preserving augmentation to training data only; use consistent preprocessing for validation and testing.
3. 2D CNN models: Train at least one CNN from scratch (e.g., a small custom CNN) and one model leveraging transfer learning (e.g., ResNet18/34) for multi-class classification.
4. Basic experimentation: Explore hyperparameters such as learning rate, batch size, and regularization (dropout, data augmentation).
5. Evaluation: Report accuracy, macro-F1, and a confusion matrix. Compare both models on the same test set and analyze common errors and class imbalance.

If you aim to score higher than 50%, pick one direction and explore in-depth:

1. Explore advanced image classification methods, such as attention or a pre-trained vision transformer. Compare against the required CNN baselines. Use ablations to explain gains from architecture, augmentation, or class balancing.
2. Investigate background, lighting, or image-quality robustness. Compare RGB, grayscale, or background-removed images using identical leaf splits, or test labeled field images with matching classes. Discuss generalization beyond controlled photographs.
3. Explore other creative ideas for plant disease classification, such as interpretability or efficient inference. Support claims with experiments and connect your idea to the course content.

Topic 2: Sentiment Analysis (Natural Language Processing)

This project focuses on building and evaluating models for sentiment analysis, a core task in Natural Language Processing. The goal is to classify business reviews as either positive or negative using the **Yelp Review Polarity dataset**, a standard benchmark for binary sentiment classification.

Dataset:

1. This topic is based on the **Yelp Review Polarity dataset**. It contains 560,000 business reviews for training and 38,000 for testing, which are labeled as either positive or negative. The dataset is available for public download at https://huggingface.co/datasets/fancyzhx/yelp_polarity.
2. Classify reviews as negative (label 0) or positive (label 1) in the Hugging Face version. A stratified subset of 20,000 training-pool reviews and 5,000 test reviews is sufficient. Report seeds and sample sizes; use the same subsets for all models.

Minimum Requirements (you secure 50% scores if the following are done correctly):

1. Data Split: Use stratified train/validation subsets of the official training split; reserve official test data for final evaluation. Record sizes and seeds.
2. Text Preprocessing & Vectorization: Clean the raw text data (e.g., normalize whitespace, convert to lowercase). Then, perform tokenization, build a vocabulary from your training data, and convert your text sequences into integer sequences. Ensure you handle sequences of varying lengths by implementing padding.
3. Deep Learning Model: Implement and train one recurrent neural network (e.g., an LSTM or GRU) for binary classification. Your implementation must include and compare the following two approaches for the embedding layer: A randomly initialized embedding layer that is trained from scratch, along with the rest of your model, and an embedding layer initialized with pre-trained word embeddings (e.g., GloVe or Word2Vec), which can be either frozen or fine-tuned during training.
4. Basic Experimentation: Explore and report on the effect of key hyperparameters for your deep learning model, such as learning rate, batch size, dropout rate, and the number of recurrent units.
5. Evaluation: Select proper evaluation metrics for classification (e.g., accuracy, precision, recall, F1-score). Analyze and compare the results you obtained from using the trainable embedding versus the pre-trained embedding.

If you aim to score higher than 50%, pick one direction and explore in-depth:

1. Explore more advanced modeling techniques by fine-tuning a large, pre-trained language model (e.g., a variant of BERT or another suitable model). You should thoroughly analyze the trade-offs in terms of performance improvement, computational cost, and implementation complexity when compared to your recurrent model.
2. Focus on a more complex NLP task with suitable labels. For example, use the Yelp Review Full dataset for 1–5 star prediction, or a separately labeled dataset for aspect-based sentiment, sarcasm, or irony. Yelp Review Polarity provides only binary labels.
3. Propose and implement any other interesting and creative ideas relevant to sentiment analysis or text classification. You could explore different model architectures, investigate methods for model interpretability to understand *why* your model makes certain predictions, or apply your models to a completely different domain, such as social media or product reviews. You must clearly explain how your chosen idea connects to the course content.

Topic 3: Open Topic

If you are interested in another AI problem that aligns with the course content, you may propose your own project. This is an opportunity to explore areas like generative AI, multimodal learning, foundation models, or other advanced deep learning applications.

- You are free to choose any relevant dataset and AI/deep learning methods.
- Your project should demonstrate a clear understanding of the principles and techniques taught in this course.
- Remember that the chosen topic's difficulty, scope, and creativity will be key factors in your evaluation.

3. Evaluation Criteria

Your project will be evaluated based on a holistic assessment of your work across several dimensions:

- **Project Completeness & Model Performance:** How thoroughly the project tasks were completed and the effectiveness of your final model(s) on the chosen task.
- **Creativity & Difficulty:** The novelty of your approach, the technical challenge of the problem, and the sophistication of the methods used.
- **Code Quality:** The readability, organization, and documentation of your source code. Clean and well-structured code is expected.
- **Storytelling (Report & Presentation):** The clarity and depth of your project report and presentation. This includes how well you explain your motivation, describe your methodology, analyze your results, and present your conclusions.